Generative Conformal Prediction with Optimized Coverage Allocation

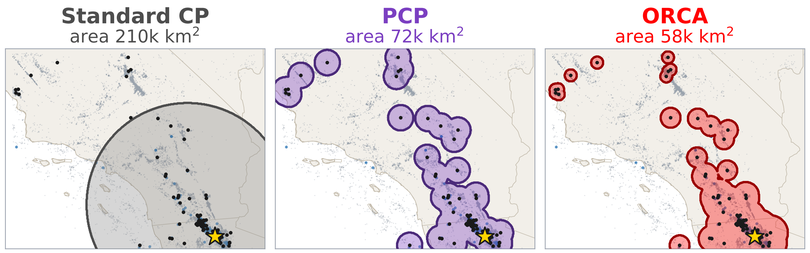
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Interactive demo

Motivation: Where Will the Next Earthquake Strike?



Conformal Prediction. CP provides an uncertainty set $\hat{C}(X_{\text{test}})$ for testing input X_{test} that contains Y_{test} with probability $1-\alpha$ that is model agnostic and distribution-free.

$$\Pr\left(Y_{\text{test}} \in \hat{C}(X_{\text{test}})\right) \;\geq\; 1-\alpha.$$

Limitation.

- Standard CPs can be inefficient in multimodal settings, over low-density spaces, and cannot utilize generative models.
- PCP takes advantage of generative models, but not adaptive to the distribution of the generated samples, leading to some inefficiency.

Problem Setup

Input. A conditional generative model $\hat{p}(Y \mid X)$ we can sample from, $Y \in \mathbb{R}^d$, a labeled dataset $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^{n_1+n_2}$, and a test input X_{test}.

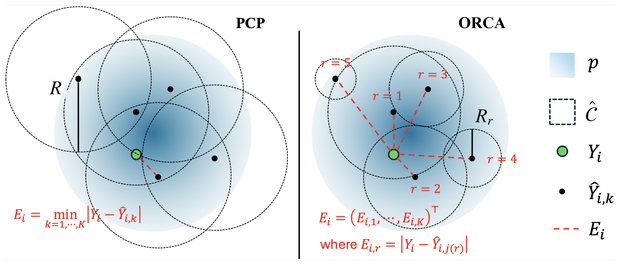
Goal. A constrained optimization: minimize set size while maintaining validity,

$$\min \left|\hat{C}(X_{\text{test}})\right| \qquad \text{s.t.} \qquad \Pr\left(Y_{\text{test}} \in \hat{C}(X_{\text{test}})\right) \;\geq\; 1-\alpha.$$

Uncertainty Set Construction. A union of K balls with adaptive radii:

$$\hat{C}(X_{\text{test}}) = \bigcup_{r=1}^{K} \left\{y : \|y - \hat{Y}_{\text{test},j(r)}\| \leq R_r^*\right\}, \qquad \text{PCP: } R_r^* = R \;\text{ for all } r.$$

Key Idea: A Ranked Score Vector



PCP. One score per point and one radius R for every sample.

ORCA. The full distance vector, ordered by density, with a radius R_r per rank.

Method

Stage 1 **Sampling and ranking (on $\mathcal{D}_{\text{explore}}$, size n_1)**

Draw $\hat{Y}_{i,1}, \ldots, \hat{Y}_{i,K} \sim \hat{p}(\cdot \mid X_i)$, rank them by average m-nearest-neighbor distance (densest first), and record

$$E_i = (E_{i,1}, \ldots, E_{i,K}), \qquad E_{i,r} = \|Y_i - \hat{Y}_{i,j(r)}\|.$$

Stage 2 **Optimal coverage allocation**

Choose a radius R_r per rank, an empirical quantile of $\{E_{i,r}\}$:

$$\min_{R_1,\ldots,R_K} \sum_{r=1}^{K} R_r^d \quad \text{s.t.} \quad \frac{1}{n_1}\sum_{i=1}^{n_1} \max_r \mathbf{1}\{E_{i,r} \leq R_r\} \;\geq\; 1-\alpha.$$

Solved exactly as a MILP, or fast by an LP relaxation or a greedy set-cover rule.

Stage 3 **Conformal calibration (on $\mathcal{D}_{\text{calib}}$, size n_2)**

$$S_i = \min_r \frac{E_{i,r}}{R_r}, \qquad R_r^* = t_\alpha R_r,$$

where t_α is the $\lceil (1-\alpha)(n_2+1) \rceil$-th smallest S_i.

Theoretical Guarantees

Theorem 1 *(exact MILP)*

The MILP is equivalent to the Stage 2 problem; its optimum is globally optimal.

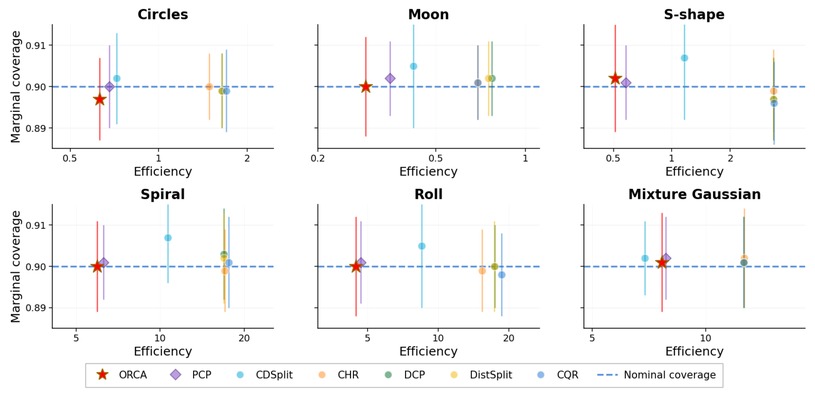
Theorem 2 *(finite-sample validity)*

If the test pair is exchangeable with $\mathcal{D}_{\text{calib}}$, then $\Pr\left(Y_{\text{test}} \in \hat{C}(X_{\text{test}})\right) \geq 1-\alpha$ for any $R_1, \ldots, R_K$, so the MILP, LP and greedy solvers are all valid.

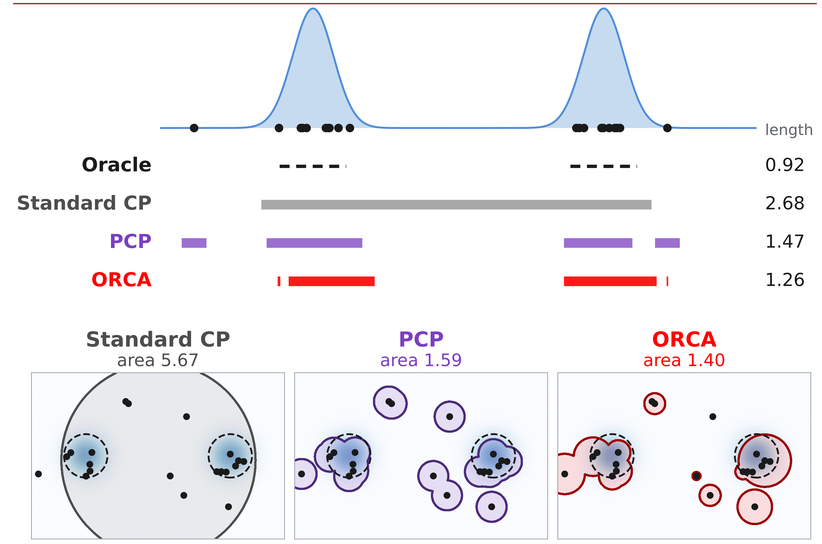
Theorem 3 *(convergence to the oracle)*

As $K, n_1, n_2 \to \infty$, $\left|\hat{C}(x) \triangle L_{\tau^\star}(x)\right| \to 0$ in probability, where $L_{\tau^\star}(x)$ is the $(1-\alpha)$ highest-density region.

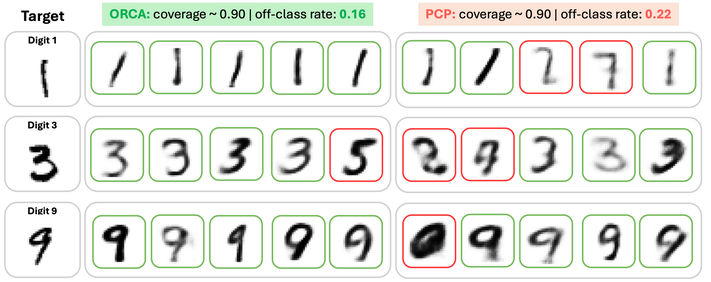
Synthetic Data: Smaller Sets, Same Coverage



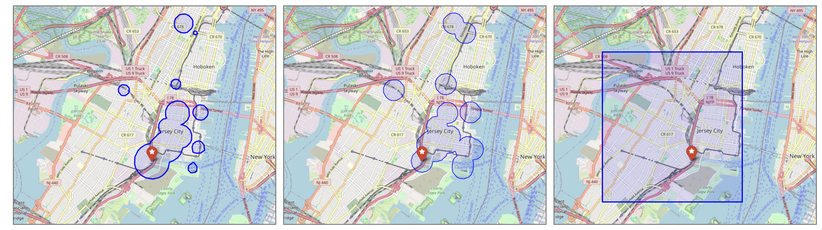
Prediction Sets on Synthetic Data



MNIST Digit Generation



NYC Bike Destinations



Map data © OpenStreetMap contributors.